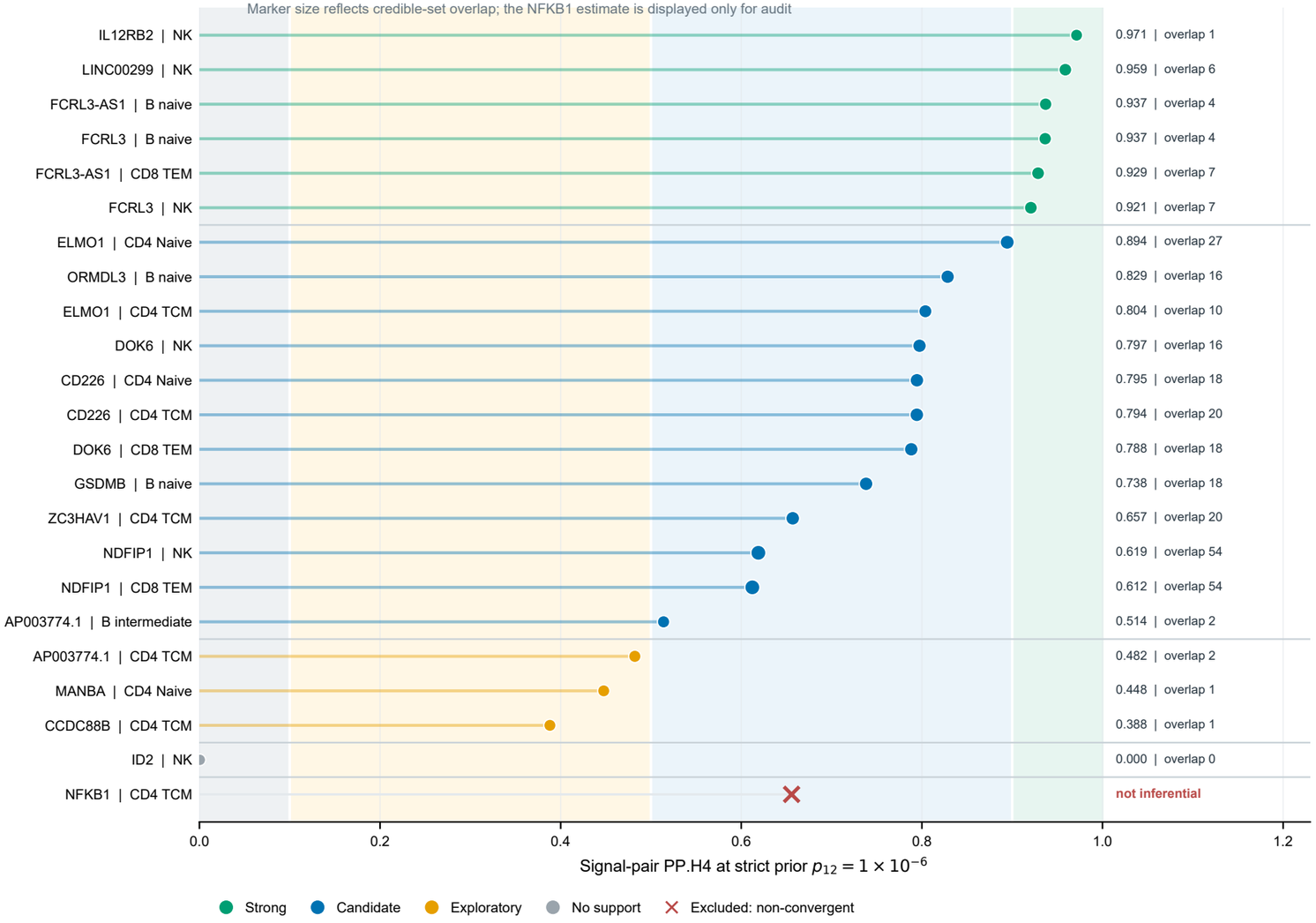
Screen-wide signal-level colocalization and model quality control

All 23 prespecified gene–cell combinations are shown; no result was removed because of statistical direction

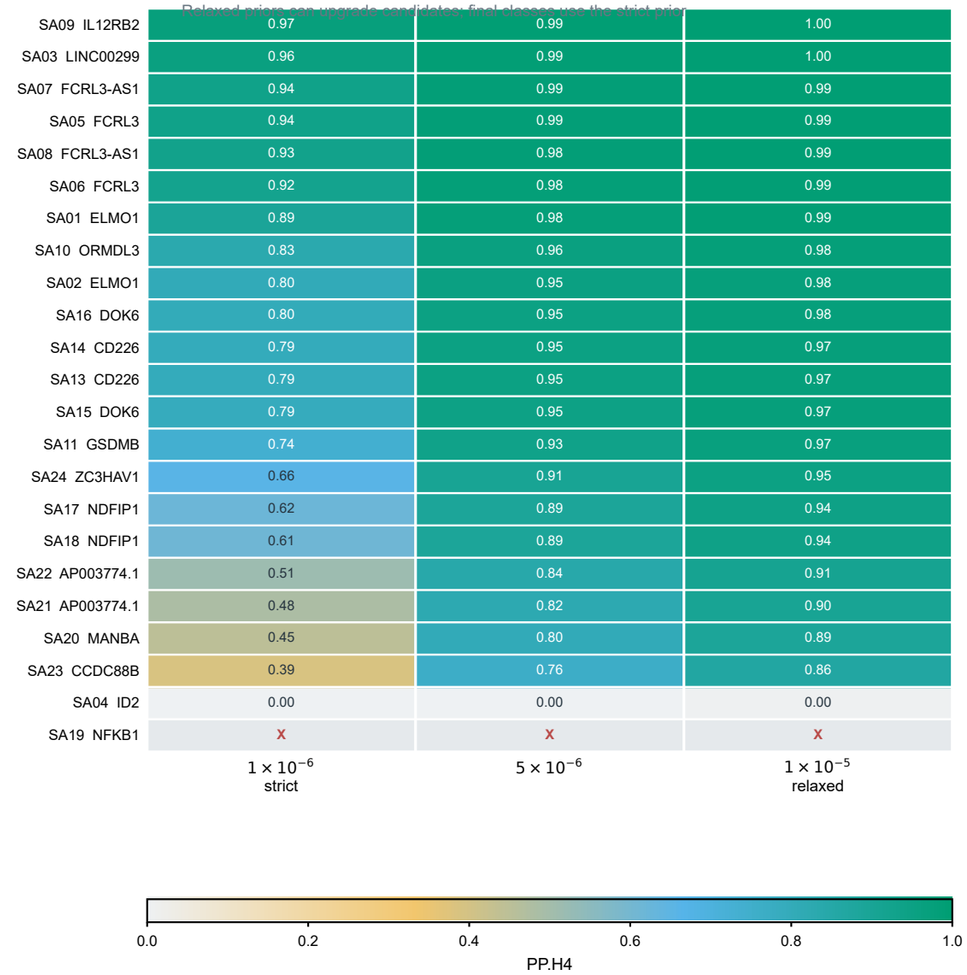
A Complete prespecified discovery screen

Marker size reflects credible-set overlap; the NFKB1 estimate is displayed only for audit



B Prior sensitivity across all combinations

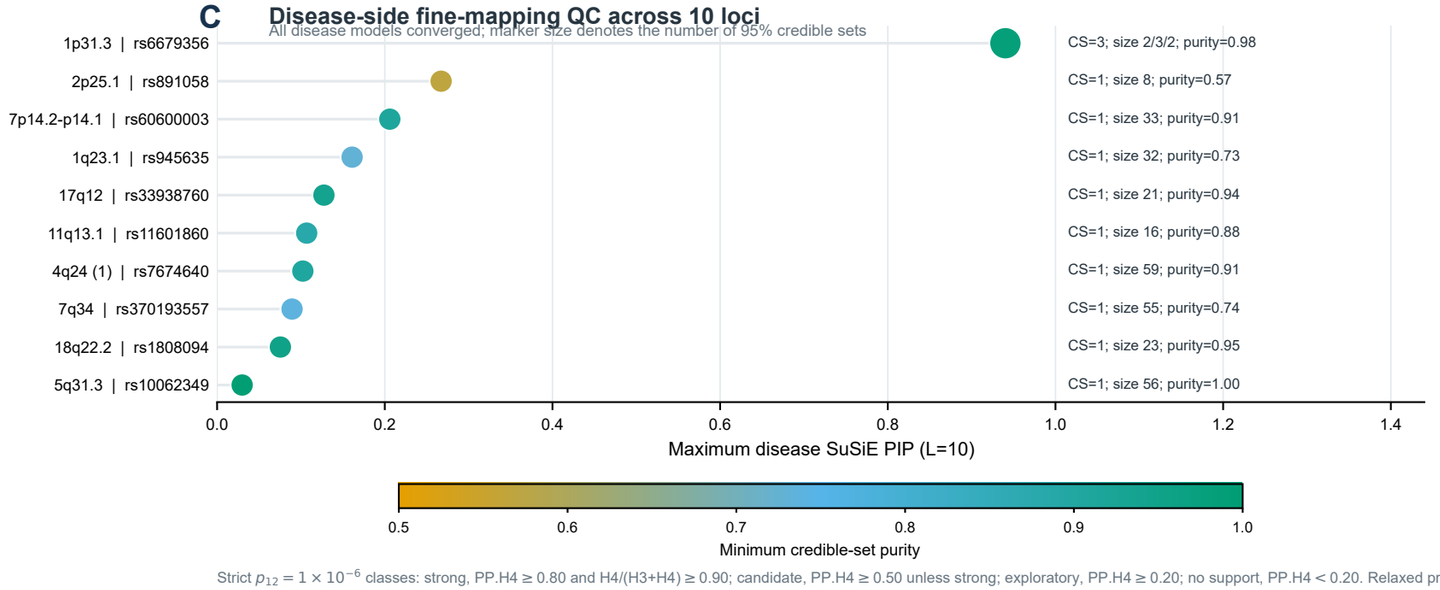
Relaxed priors can upgrade candidates; final classes use the strict prior



NFKB1 values arise from a non-convergent eQTL model and are crossed out rather than interpreted.

C Disease-side fine-mapping QC across 10 loci

All disease models converged; marker size denotes the number of 95% credible sets



D Convergence audit and prespecified exception handling

A failed model is excluded unless a lower-complexity review converges

Disease SuSiE 20 / 20 L=3 and L=10 models	**eQTL SuSiE L=10** 21 / 23 two models reviewed at L=3
L=3 review 1 / 2 IL12RB2 rescued; NFKB1 failed	**Final usable** 22 / 23 one model excluded

SA09 IL12RB2 / NK L10 failed → L3 converged → INCLUDED

SA19 NFKB1 / CD4 TCM L10 failed → L3 failed → EXCLUDED

Interpretation lock: PP.H4 from a non-convergent model is an audit trace, not biological evidence.

Strict $p_{12} = 1 \times 10^{-6}$ classes: strong, PP.H4 ≥ 0.80 and H4/(H3+H4) ≥ 0.90; candidate, PP.H4 ≥ 0.50 unless strong; exploratory, PP.H4 ≥ 0.20; no support, PP.H4 < 0.20. Relaxed priors cannot upgrade a class; same-locus combinations are not independent discoveries.